WORKED SOLUTION

STEP III 2025, Question 9

Mechanics • Centre of Mass

STEP III 2025, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 11.8, *Tilted Equilibria*: https://beyondstep.keithmcnulty.org/ch11/tilted-equilibria.html

The key idea

A solid resting on a table touches it at one point, and only two forces act: the weight at the centre of mass G and the reaction at the contact point. They can balance only if they act along the same vertical line, so the normal to the surface at the contact point must pass through G. Part (i) locates G, part (ii) tells us where normals to the curved surface meet the axis, and part (iii) puts the two together: we count the points on the surface whose normals pass through G.

Solution

Throughout, write the curve as $y = \dfrac{r^n - x^n}{r^{n-1}}$ for $0 \le x \le r$. It runs from $(0, r)$ on the y-axis down to $(r, 0)$ on the x-axis, with $y \ge 0$ in between.

Part (i)

We cut the solid into thin cylindrical shells about the y-axis. The shell at radius x, of thickness δx and height y, has volume $2\pi x y\,\delta x$, and since the density is uniform its centre of mass is at height $\frac{1}{2}y$. So the volume and the moment about the xz-plane are

$$V = 2\pi \int_0^r xy \,\mathrm{d}x, \qquad V\bar{y} = 2\pi \int_0^r x \cdot \tfrac{1}{2}y^2 \,\mathrm{d}x.$$

The first integral is

$$\int_0^r x\,\frac{r^n - x^n}{r^{n-1}}\,\mathrm{d}x = \frac{1}{r^{n-1}}\left(\frac{r^{n+2}}{2} - \frac{r^{n+2}}{n+2}\right) = \frac{n\,r^3}{2(n+2)},$$

so $V = \dfrac{\pi n r^3}{n+2}$. For the second, expand $(r^n - x^n)^2$:

$$\int_0^r x\,\frac{(r^n - x^n)^2}{r^{2n-2}}\,\mathrm{d}x = \frac{r^{2n+2}}{r^{2n-2}}\left(\frac{1}{2} - \frac{2}{n+2} + \frac{1}{2n+2}\right)$$
$$= r^4 \cdot \frac{(n+2)^2 - 4(n+1)}{2(n+1)(n+2)} = \frac{n^2 r^4}{2(n+1)(n+2)},$$

so $V\bar{y} = \pi \cdot \dfrac{n^2 r^4}{2(n+1)(n+2)}$. Dividing,

$$\bar{y} = \frac{\pi n^2 r^4}{2(n+1)(n+2)} \cdot \frac{n+2}{\pi n r^3} = \frac{nr}{2(n+1)},$$

as required. By symmetry the centre of mass lies on the y-axis.

Part (ii)

Differentiating, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{nx^{n-1}}{r^{n-1}}$, which at $x = rp$ is $-np^{n-1}$. This is non-zero since $p > 0$, so the normal has gradient $\dfrac{1}{np^{n-1}}$ and equation

$$y - r(1 - p^n) = \frac{x - rp}{np^{n-1}}.$$

Putting $x = 0$,

$$Y = r(1 - p^n) - \frac{rp}{np^{n-1}} = r\left(1 - p^n - \frac{1}{np^{n-2}}\right),$$

which is the required result.

For $n = 4$, $Y(p) = r\left(1 - p^4 - \frac{1}{4}p^{-2}\right)$ and

$$\frac{\mathrm{d}Y}{\mathrm{d}p} = r\left(-4p^3 + \frac{1}{2p^3}\right) = \frac{r(1 - 8p^6)}{2p^3}.$$

Ask yourself How do we know the stationary point gives the greatest value, and not just a local one?

For $p > 0$ the sign of $\mathrm{d}Y/\mathrm{d}p$ is the sign of $1 - 8p^6$: positive for $p^6 < \frac{1}{8}$ and negative for $p^6 > \frac{1}{8}$. So Y increases up to $p^6 = \frac{1}{8}$, that is $p^2 = \frac{1}{2}$, $p = 1/\sqrt{2}$ (which lies in $(0, 1]$), and decreases after it. That makes it the greatest value on the whole interval:

$$Y_{\max} = r\left(1 - \tfrac{1}{4} - \tfrac{1}{4} \cdot 2\right) = \tfrac{1}{4}r.$$

Part (iii)

The two pieces. The upper boundary $r^3 y = r^4 - x^4$ is the curve of part (i) with $n = 4$, so the top piece (above $y = 0$) is the solid of part (i) with $n = 4$. The lower boundary

$ry = -(r^2 - x^2)$ is the reflection in the x-axis of $ry = r^2 - x^2$, the $n = 2$ curve, so the bottom piece is the $n = 2$ solid turned upside down. The two meet in the circular rim of radius r in the plane $y = 0$. Using part (i), with $V = \pi n r^3/(n+2)$:

	volume	height of centre of mass
top ($n = 4$)	$\frac{2}{3}\pi r^3$	$\frac{4r}{10} = \frac{2}{5}r$
bottom ($n = 2$, reflected)	$\frac{1}{2}\pi r^3$	$-\frac{2r}{6} = -\frac{1}{3}r$

Taking moments (assuming, as before, uniform density),

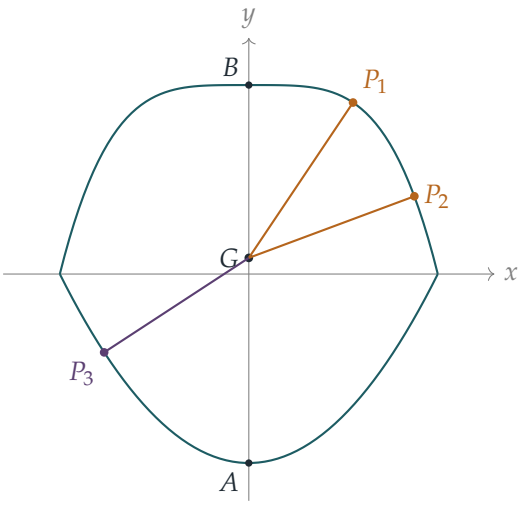
$$\bar{y} = \frac{\frac{2}{3}\cdot\frac{2}{5}r - \frac{1}{2}\cdot\frac{1}{3}r}{\frac{2}{3}+\frac{1}{2}} = \frac{\frac{1}{10}r}{\frac{7}{6}} = \frac{3r}{35},$$

so $G = \left(0, \frac{3}{35}r\right)$, a little above the rim.

The equilibrium condition.

? **Ask yourself** Why must the normal at the point of contact pass through G?

Resting on the surface at a single point P, the solid feels only its weight, acting vertically at G, and the reaction from the surface, acting at P. Two forces balance only if they are equal, opposite and along the same line, so G must be vertically above P. The surface is the tangent plane at P, so the vertical through P is the normal to the solid there: the normal at P must pass through G. Conversely, if it does, the solid can rest there, because the solid is convex (the profile $y = r - x^4/r^3$ is concave and $y = (x^2 - r^2)/r$ is convex), so it lies entirely on one side of its tangent plane at P. The normal to a surface of revolution lies in a plane through the axis, so we only need to look at the cross-section in the xy-plane. Since G is on the axis, we need normals that meet the y-axis at $Y = \frac{3}{35}r$.



Contact on the top surface. By part (ii) with $n = 4$ (and $r(1-p^4)$ the height of the point), the normal at $\big(rp, r(1-p^4)\big)$ meets the axis at $r\big(1-p^4-\frac{1}{4}p^{-2}\big)$. Setting this equal to $\frac{3}{35}r$ and multiplying by $140p^2$:

$$140p^2 - 140p^6 - 35 = 12p^2 \iff h(q) := 140q^3 - 128q + 35 = 0, \quad q = p^2.$$

Now $h(0) = 35 > 0$, $h(\frac{1}{2}) = 17.5 - 64 + 35 = -11.5 < 0$ and $h(1) = 47 > 0$. So h has a root $q_1 \in (0, \frac{1}{2})$ and a root $q_2 \in (\frac{1}{2}, 1)$, giving two points P_1, P_2 on the top surface with $0 < p_1 < 1/\sqrt{2} < p_2 < 1$. (This is what part (ii) predicts: $\frac{3}{35}r$ is below the greatest value $\frac{1}{4}r$, so the level is reached once on each side of the maximum.)

Contact on the bottom surface. For the $n = 2$ curve the normal at parameter p meets the axis at $r(1-p^2-\frac{1}{2}) = r(\frac{1}{2}-p^2)$; reflecting in the x-axis, the normal to the bottom surface at $\big(rp, -r(1-p^2)\big)$ meets the axis at $r(p^2-\frac{1}{2})$. Setting this equal to $\frac{3}{35}r$ gives

$$p^2 = \tfrac{1}{2} + \tfrac{3}{35} = \tfrac{41}{70}, \qquad p_3 = \sqrt{41/70} \in (0,1),$$

a third point P_3 (taking the positive root, since $p > 0$ on the half-section we are using; the figure shows its mirror image, which is equivalent by symmetry).

The three angles are different and non-zero. When the solid rests at P, the upward vertical points from P towards G, that is, along the inward normal at P.

- At P_3 on the bottom, the inward normal $(-2p_3, 1)$ (the curve has gradient $2p_3$) points upwards, making the acute angle $\arctan(2p_3)$ with $\overrightarrow{AB}$, which points along the positive y-axis.
- At P_1 or P_2 on the top, the inward normal $(-4p^3, -1)$ points downwards, so the angle with $\overrightarrow{AB}$ is the obtuse angle $180^\circ - \arctan(4p^3)$.

Since $p_1 \neq p_2$ and $\arctan(4p^3)$ is strictly increasing, the two obtuse angles differ, and both differ from the acute one. None is zero, because $p_1, p_2, p_3 > 0$. So the solid can rest in equilibrium with $\overrightarrow{AB}$ at three different non-zero angles to the upward vertical, as required.

(There are in fact more: the solid can also balance on its rim, or on B with $\overrightarrow{AB}$ pointing straight down. The three curved-surface positions are enough for the question.)

Checks

- Part (i) with $n = 2$ gives $\frac{1}{3}r$ for a paraboloid of height r, the standard result that a solid paraboloid's centre of mass is two-thirds of the way from its vertex to its base. As $n \to \infty$ the solid becomes a cylinder of height r, and $\bar{y} \to \frac{1}{2}r$, as it should.
- In part (ii), at $p = 1$ (the rim) with $n = 4$ we get $Y = -\frac{1}{4}r$; the gradient there is -4, so the normal has gradient $\frac{1}{4}$ and drops $\frac{1}{4}r$ over a horizontal distance r. It agrees.

- With $r = 1$ the roots are $p_1 \approx 0.552$, $p_2 \approx 0.876$, $p_3 \approx 0.765$, and the angles are about 146.1°, 110.4° and 56.8°. A computer search for the tilts at which the height of G above the table is stationary found exactly these three strictly between 0° and 180°, together with the rim position (about 85.1°).

Common slips

- Placing the bottom piece's centre of mass at $+\frac{1}{3}r$: it is the $n = 2$ solid turned upside down, so its centre of mass is below the rim.
- Using disc volumes $\pi x^2\,\delta y$ with x still in terms of y to the power $\frac{2}{n}$: it works, but the shell method gives polynomial integrals.
- Finding the cubic in $q = p^2$ but not justifying that it has two roots with $0 < p < 1$; a sign check at $q = 0, \frac{1}{2}, 1$ settles it.

Beyond STEP

The condition that the normal passes through G makes the height of G stationary as the body rolls, and that height is the potential energy. Stable positions are its minima and unstable ones its maxima. A smooth convex body must have at least two equilibria, and the "Gömböc", discovered in 2006, is a convex shape with only one stable and one unstable position. The general machinery of potential energy and stability is in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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